

Decision variables

feedstock chemistry (CaO + MgO mass fractions) · grain size (10–500 μm) · allocation rule (LiTAS-targeted, uniform 10/20/50 t ha^{-1}) · carbon price ($\$/\text{tCO}_2$) · MRV ($\$/\text{tCO}_2$) · discount rate (NPV)

Biogeochemistry

CDR stream $\rightarrow R_{\text{CDR}}$

Feedstock $\times$ grain size
reactive Ca/Mg surface from
CaO, MgO and SSA exponent

Dissolution kinetics
 $f(\text{MAT}, \text{MAP}, \text{pH}) - \text{Ar-}$
 $\text{henius} \times \text{acid catalysis}$

Net alkalinity export
total CDR potential – pedogenic-
C precipitation (aridity)

$$R_{\text{CDR}} = \text{CDR}_t \times (\text{pCO}_2 - \text{MRV})$$

Agronomy

yield-uplift stream $\rightarrow R_{\text{agro}}$

Soil acidity constraint
pH, exchangeable acidity,
Al saturation, ECEC

sets rate

Lime-eq demand $\rightarrow$ basalt rate
Kamprath / Cochrane /
Aramburu-Merlos (LiTAS)

Crop response
EcoCrop $\cup$ Bayesian uplift
(trial-anchored posterior)

$$R_{\text{agro}} = \Delta y(x) \times p_{\text{crop}}$$

Economic geography

cost stream $\rightarrow C$

Feedstock supply
basalt fines at quarry
gate; GLiM source rocks

Logistics
cost-distance over friction
surface $\times$ $\$/\text{min/t}$

Grinding (cost + LCA)
 $\text{kWh/t} \times \text{country tariff} \times \text{grid CI}$

$$C = \text{supply} + \text{logistics} + \text{grinding} + \text{spreading} + \text{MRV}$$

Per-pixel gross margin

$$\pi(x, y \mid \text{regime, allocation}) = R_{\text{CDR}}(x, y) + R_{\text{agro}}(x, y) - C(x, y)$$

regimes: year-1, NPV @ 10%, equilibrium • allocations: LiTAS-targeted, uniform 10/20/50 t ha^{-1}

Profitability map

$\$/\text{ha/yr} - 23 \text{ crops} \times$
 $4 \text{ alloc.} \times 3 \text{ regimes}$

Supply curves

rank pixels by $\pi/\text{t basalt}$
 $\rightarrow \text{MtCO}_2 @ \$/\text{tCO}_2$

Sensitivity sweeps

price $\times$ yield $\times$ MRV
 $\times$ grain $\times$ allocation